

THE Ohio Naturalist

Volume II.
Number 2.

DECEMBER, 1901.

Annual Subscription, 50 cts.
Single Number, 10 cts.

“The Best of Everything Laboratorial.”

For every division of Natural Science.

Microscopes and Accessories.

Slides and Cover Glasses.

Pocket Magnifiers.

Field Glasses.

Dissecting Instruments.

Insect Pins.

Glassware,

Chemicals,

Etc., Etc.

General Apparatus & Chemical Company

Cleveland, Ohio, 262 Prospect St.

Branch: Berlin, Germany.

THE
Ohio Naturalist

Volume II.
Number 2.

DECEMBER, 1901.

Annual Subscription, 50 cts.
Single Number, 10 cts.

“The Best of Everything Laboratorial.”

For every division of Natural Science.

Microscopes and Accessories.
Slides and Cover Glasses.
Pocket Magnifiers.
Field Glasses.
Dissecting Instruments.
Insect Pins.
Glassware,
Chemicals,
Etc., Etc.

General Apparatus & Chemical Company

Cleveland, Ohio, 262 Prospect St.

Branch: Berlin, Germany.

Entered at the Post Office at Columbus, Ohio, as second class matter.

The Project Gutenberg eBook of The Ohio Naturalist, vol. II, no. 2, December, 1901

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The Ohio Naturalist, vol. II, no. 2, December, 1901

Creator: Ohio State University. Biological Club

Release date: August 31, 2023 [eBook #71531]

Language: English

Original publication: Columbus, OH: The Biological Club of the Ohio State University, 1900

Other information and formats: www.gutenberg.org/ebooks/71531

Credits: Richard Tonsing and the Online Distributed Proofreading Team at <https://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive)

*** START OF THE PROJECT GUTENBERG EBOOK THE OHIO
NATURALIST, VOL. II, NO. 2, DECEMBER, 1901 ***

Transcriber's Note:

New original cover art included with this eBook is granted to the public domain.

THE
Ohio Naturalist

Volume II.

**DECEMBER,
1901.**

Annual Subscription, 50 cts.

Number 2.

Single Number, 10 cts.

“The Best of Everything Laboratorial.”
For every division of Natural Science.

Microscopes and Accessories.
Slides and Cover Glasses.
Pocket Magnifiers.
Field Glasses.
Dissecting Instruments.
Insect Pins.
Glassware,
Chemicals,
Etc., Etc.

General Apparatus & Chemical Company

Cleveland, Ohio, 262 Prospect St.

Branch: Berlin, Germany.

Entered at the Post Office at Columbus, Ohio, as second class matter.

....LEARN....

Bookkeeping, Shorthand and Typewriting

at the

Ohio Business
Institute....

Y. M. C. A. Bldg.
Columbus, Ohio



Bookkeeping taught here as books are kept. All entries made from vouchers, bills, notes, checks, drafts, etc. The student learns by doing. Thorough, interesting and practical.

We are pioneer teachers of **Gregg's Light-Line Shorthand**, a simple, sensible, legible, rapid system. Learned in half the time required by other systems. Send postal for first lesson.

If you are at present employed, write us regarding our Home

Study work.

Same thorough instruction as given in the class room.

Circulars and Booklets free to those desiring particulars concerning an up-to-date School of Business.

ROWLAND & ROGERS, Proprs.,
Columbus, O.

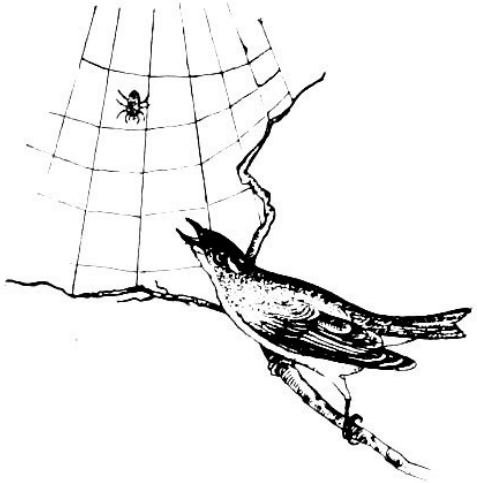
K. C. R. L. N. M. T. D. A. E. I.
WRITE BY SOUND: (up) o o o
me meek get day cat near
met make tray cake inv keg
TO BE MEMORIZED:
A, to Can Good in He The Will Period
WRITE:
The lad will meet me in the lane.
READ:

A LESSON IN GREGG'S
SHORTHAND.

Scientific Taxidermy.

EVERY SCHOOL should have a collection of Natural History specimens.
Why not

START A COLLECTION



of the *Birds* or *Mammals* of your county or state, or enrich a present collection by the addition of some well mounted specimens? Costs so much? Perhaps so, but WRITE TO ME FOR PRICES and you will be surprised how little such a collection will cost after all. Think of the interest that will be awakened in your school in the

STUDY OF NATURE

to reward you for trouble and outlay.

I mount to
order

BIRDS, MAMMALS, FISH and REPTILES

and solicit your patronage. If you are interested, you should not lose an opportunity to examine the novel collection of Ohio birds, prepared by myself, in the museum of Zoology, Ohio State University.

TAUGHT BY MAIL.—I give instructions in the Art of Taxidermy, personally, or BY MAIL. You can learn to collect and prepare your own birds, thus reducing the cost of a collection to a minimum.

For prices and circulars, address

NATURALIST—THOMAS M. EARL—
TAXIDERMIST



Cor. Naghten and Ninth Sts., COLUMBUS, OHIO.

The "Ohio Naturalist."

A journal devoted more especially to the natural history of Ohio. The official organ of THE BIOLOGICAL CLUB OF THE OHIO STATE UNIVERSITY. Published monthly during the academic year, from November to June (8 numbers). Price 50 cents per year, payable in advance. To foreign countries, 75 cents. Single copies, 10 cents.

Editor-in-Chief, JAMES S. HINE.

Associate Editors.

J. A. BOWNOCKER, Geology,
J. H. SCHAFFNER, Botany,
F. L. LANDACRE, Zoology,
W. C. MILLS, Archaeology,
MAX MORSE, Ornithology,
F. J. TYLER, Ecology.

Advisory Board.

Prof. W. A. KELLERMAN.
Prof. HERBERT OSBORN.
Prof. CHARLES S. PROSSER.

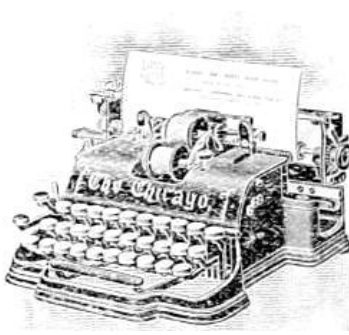
Address **THE OHIO NATURALIST**,
Ohio State University, Columbus, Ohio.

The Truth, The Whole Truth,

and Nothing but the Truth.

UNCONDITIONALLY THE BEST TYPEWRITER IS

THE CHICAGO \$35 00



No typewriter is worth \$100. We have made a mechanically **excelling** machine, and sell it for \$35. We claim that it is **the superior** of any typewriter made. This is a broad but carefully weighed statement, and it is the **truth**.

Awarded gold medal at the Paris Exposition, 1900, in open competition with all other makes of typewriters.

Our descriptive matter tells an interesting story. Send for it and learn something about a high-grade typewriter sold at an honest price.

CHICAGO WRITING MACHINE CO.,
89 Wendell Street,  CHICAGO, U. S. A.

The Ohio Naturalist,

PUBLISHED BY

The Biological Club of the Ohio State University.

Volume II.

DECEMBER, 1901.

No. 2.

TABLE OF CONTENTS.

TYLER—Meeting of the Biological Club	147
Meeting of the Ohio State Academy of Science	156
KELLERMAN—Fifty Additions to the Catalogue of Ohio Plants	157
KELLERMAN—Botanical Correspondence, Notes and News for Amateurs, I	159
KELLERMAN—Note and Correction to Ohio Fungi Exsiccate	161
GRIGGS—Notes of Travel in Porto Rico	162
MORSE—Salamanders Taken at Sugar Grove	164
WILLIAMSON—Fishes Taken Near Salem, Ohio	165
HINE—Collecting Tobanidae	167
HINE—Observations on Insects	169

MEETING OF THE BIOLOGICAL CLUB.

November 4th, 1901.

The Biological Club met in Orton Hall and was called to order by the president, Prof. Osborn. As it is customary to elect new officers at the November meeting each year, the Nominating Committee presented the following names: For president, Mr. Mills; for vice-president, Mr. Morse; for secretary, Mr. Tyler. Prof. Lazenby moved that the secretary be instructed to cast the unanimous ballot of the members present for the names proposed. Carried. Messrs. J. C. Bridwell, M. T. Cook and Harvey Brugger were elected members.

The retiring president, Prof. Osborn, presented a very interesting address, an abstract of which follows:

SCOPE OF MODERN BIOLOGY.

It has been the custom in this society, following a mandate of its constitution, for the president on retiring from the chair to give an address, and it is presumed that such an address will either bring to your attention the results of some special investigation, summarize the work in some field of research or outline the progress and problems with which biology has to do.

When a year ago you were so kind as to honor me with this office, two things I think came especially to my mind; one the success of the club particularly in the new enterprise of publishing a journal; the other the duty, honor and privilege of preparing an address for this occasion. I presume you have all had the experience of contemplating some distance in the future a certain duty, debating the most suitable theme or method, and perhaps seen the time grow shorter and shorter with little real accomplishment. If I were to enumerate the various topics that have come to my mind as suitable for this occasion it would exhaust quite a part of our time; if I could reproduce the current of thought that has flowed from time to time along the pathways of such topics, I am sure you would experience a weariness that I should regret to occasion.

The parts of biology which we may make thoroughly our own are very few. It may be profitable, therefore, occasionally to take a general survey of the field to see what its sphere of influence may be, what phases of life are being advanced by its discoveries or by the distribution of knowledge which follows. It has seemed to me therefore that it would be appropriate this evening to attempt some such survey of biology, even though it be fragmentary and inadequate.

For convenience in arrangement we may group this survey along the lines of practical applications of service to mankind, such as occur in medicine, agriculture and kindred industries, domestic and social life, and those which have to do with the acquisition of knowledge and with education.

Applications of biology in medical science, in agriculture and in domestic life have in many cases assumed such intimate and essential

character that we often look upon them as applied sciences more than in any other way.

While biology has been the foundation of all rational systems of medicine and the constant servant of this most beneficent of human professions, the forms of its uses and the wide reach of its service have so increased in recent years that we almost have excuse in feeling that it is a modern acquisition.

Could the ancient disciples of Esculapius, with their views of physiology and anatomy, have seen the present scope of these subjects and the marvelous results in cure and control of diseases by the discoveries and applications in bacteriology, I doubt if they would have recognized it as any part of their biology. Still harder would it have been to appreciate the relations of malarial parasite, mosquito and man whereby a serious disease in the latter is occasioned. Intimate relations of two kinds of life, as evidenced in the common parasites, must have been familiar from early times and their effects duly recognized, though their means of access and necessary life cycles were long misunderstood. But such relations as are found to exist in the production of malaria, Texas fever and yellow fever have been so recently discovered that we count them among the triumphs of our modern science. Indeed the discovery of such a relationship may be considered as having been impossible until the methods of modern research and the basis of knowledge as to life conditions were acquired, and which made it possible to put the disjointed fragments together. With the fragments thus related the riddle seems so simple that we wonder it was not solved before, but we must remember that it is knowledge which makes knowledge possible.

These direct advantages in medical science are however but part of the great gift to modern methods of disease control, for the possibilities in the control of disease by sanitation, quarantine, vaccination, etc., and other methods are all based on biological data.

In speaking of these recent acquisitions I would not disparage those important, in fact essential subjects of longer growth. Modern medicine would be a fragile structure without its basis of comparative anatomy, physiology, materia medica and therapeutics, which have for long years furnished a basis for rational methods in surgery and medication.

With all this knowledge at hand it is grievous to observe how general the delusion that disease may be eradicated by some much emblazoned nostrum, that some vile 'Indian compound' will be thought to have more virtue than the most accurately proportioned prescription which represents the best that modern science can do in the adaptation of a particular remedy to a particular ailment. That the patent medicine business is a most gigantic fraud and curse will I believe be granted by every scientific man who has made himself acquainted with the subject. Its immense profits are attested by the square miles of advertisements that disgrace the modern newspaper and magazine. Fortunes made from the fortunes spent in such advertising, along with the commissions to the lesser dealers, are drawn from a credulous people who not only receive no value in return, but in most cases doubtless are actually injured as a result.

That no student of biology can be deluded by such preposterous claims as characterize these compounds, in fact by any system of cure not based on sound biological principles, seems only a logical result of his training. I do not recall ever seeing the name of a biologist among the host of those who sing the praises of some of these rotten compounds. Mayors, congressmen, professors, clergymen and other presumably educated parties appear along with the host of those who fill this guilty list, a list that should be branded as a roll of dishonor. I believe that educated men owe some measure of effort toward the abatement of this plague. Naturally the medical profession is thought to be the rightful source for action, but among the uninformed any effort there is attributed to selfish motive. Certainly some measure of reform in this direction would be a service to mankind, and while no sensational crusade may be necessary, each one who knows enough of the laws of life to appreciate the monstrous folly of this business has it in his power to discourage it within the sphere of his individual influence at least. Newspapers are mostly choked off by the immense revenue derived from advertising, in fact I have known some which depended upon this as their main source of support, and have heard the candid statement that they could not have existed without it. All the more honor therefore to the few, and there are a few, which absolutely refuse to allow such advertisements in their columns.

That the modern physician must have a thorough knowledge of biology has become more and more apparent. He has to deal with life, and life thus far at least cannot be rendered into mere mechanical, physical or chemical

factors. The activities of the human machine have much that must be studied from the basis of organic nature. If we do not know all the factors or forces of life we do know that there is a complex or combination of forces radically different from any single force of inorganic nature. Chemical affinity, physical attraction and repulsion, mechanical forces may furnish many aids, but the study of life activities must go still further. To do this we must recognize the laws of organic life, the forces of growth and nutrition, of reproduction, of evolution, in fact a host of forces which have no counterpart in the inorganic world.

Modern agriculture and horticulture are so dependent on the principles of biology that to dissociate them does violence to thought. Indeed this relation has existed through all recorded history, but in no period has the utility of biologic laws been so intimately blended with all the processes of cultivation.

The determination of the zones of greatest productivity for different crops, their soil requirements, the introduction and acclimatization of species belonging to other faunal or floral regions, the essentials of animal and plant nutrition, the control of disease or abatement of noxious forms of plant or animal, all these and more are embraced in the service of biologic science to agriculture in its various forms and thus to human interests.

Among special cases cited, but which cannot be printed here in detail, were various plant diseases, and particularly various insect pests, and the discoveries which have brought them more or less under control.

Aside from the sources of food supply, which come under the general term of agriculture, we derive many articles of diet from sources dependent on animal or plant life. The various fishery industries and oyster culture which have been so wonderfully promoted by biological investigations are excellent examples of the service of science to mankind. Game laws for the protection of certain forms of life of utility to man and the possible sources of food from various animals or plants not yet utilized may be mentioned here. Clothing comes in for its share, as in the methods for protection of silkworms, the saving of fur seals and other fur-bearing animals from extinction, and the use of various fibre plants. The successful growth of sponges, of pearls and many other articles of domestic comfort or ornament are connected in one way or another with biological problems, and their

fullest development dependent on rational measures possible when the biological conditions are known.

In another way these questions enter into our social and commercial life. The rights of property in the migrant or semi-migrant forms of life have biologic as well as legal basis and some quite peculiar legal decisions would doubtless have been very different had the biology been appreciated. The classification of turtles as 'vermin' since they are neither fish nor fowl may be given as a case in point. Equally absurd and sometimes more disastrous are some of the rulings by customs officers whose knowledge of biology was doubtless derived from a Greek lexicon or some equally good authority. Such quarantine restrictions as have been imposed upon certain products by some governments show total lack of knowledge as to the possible conditions of injurious transportation or else the misapplication of them to serve some special end.

The exclusion of American pork and American fruits from certain countries, the controversy over the fur seals in Alaska, the inconsistent laws of states or nations regarding game, are some of the instances where it is evident that the law-making power and the agents of diplomacy need to be re-enforced with definite biological knowledge.

But there is another phase quite distinct from the purely utilitarian. Biological science opens up to us the facts of life and solves some of the questions of the greatest interest to mankind. What is life? What its origin? What are the factors that have controlled its development and the wonderful complexities which we observe in its distribution and adaptations? Are the forces that operate in the living organism merely physical, mechanical and chemical or are there activities inherent in life itself or that operate only in the presence of the life containing complex? Certainly, in no other branch of science are there problems more inviting. In no other has present knowledge given greater inspiration or greater intellectual service to mankind.

The field for acquisition of knowledge widens with each new discovery. We no sooner gain foothold in some hitherto unexplored realm than we become conscious that beyond this lie still other realms, knowledge of which has been dependent on knowledge of the routes by which they may be reached.

Thus structure must be known to understand function, and function known enables us to interpret structure. Evolution could not be demonstrated until after there had been gathered the necessary materials to show relations of different organisms, past and present. But, evolution known, and vast arrays of structure become intelligible. Without the knowledge of organic distribution no laws of distribution could be framed, but without the explanation of distribution afforded by evolution the facts are an unmeaning puzzle. So, too, without an effort at systematic arrangement of plant and animal forms no fundamental law of relationship could have been discovered, but given a law of relationship and systematic biology assumes a totally different aspect. Recognition of the multitudinous forms of nature are but one step then in the presentation of the vast concourse in their proper relations.

No doubt biologists will persist till every form of life has been adequately described and some means of designating it adopted. So much may be expected from the enthusiasm of the systematist. Some centuries of effort must, of course, be expected to elapse before the task is done. But it is evident that the modern biology is much less concerned in the mere recognition of these innumerable forms of life, these remotest expressions of the force of evolution, than in the gaining of some adequate conception of their relations, the forces of adaptation that have fitted them for their particular niche in the realm of nature, their relation to the other organisms with which they are associated and which constitute for them a source of support or a menace to existence. That is, modern biology concerns itself not only with the elements of structure in the organism, with the means it has of performing its varied functions with the aggregate of individuals which constitute its species, but goes on to its relations to all the influences and forces which have made it what it is and which sustain its specific existence. Less than this is too narrow a view of the province of biology. Here is unlimited scope for the student who pursues knowledge for love of knowledge.

As an inspiration to the general student the field of biology has always held an important place, and in these modern times its fascination is as potent as ever. Men have attacked the problems of life from many different viewpoints with greatly different aim and great difference in preparation and method in their work. Some of these have sought merely for inspiration for literary effort, but so far as their records have been exact and truthful

they are contributions to science, when mixed with “vain imaginings” they become literature and not science, although their right to rank here may depend on literary merit. Every gradation from pure fiction to pure science may be found and every grade of literary merit as well. White and Goldsmith, Wood and Figuiet, Kipling and Seton-Thompson, with many others that could be cited, illustrate this wide divergence among writers who have written to the entertainment and the greater or less profit of their readers. The value of such works as these is rather hard to estimate, especially from the scientific standpoint and particularly when one is under the hallucination of a beautiful piece of literary creation. They furnish entertainment and cultivate imagination, some of them stimulate observation and awaken an interest in nature, but unfortunately many of them contain so much that is inexact or erroneous that they may sadly encumber the minds of their readers.

But I would like to call attention here to what appears to me a fundamental condition of scientific work and thereby a necessary result of scientific training. Science is naught if not exact. Accurate observation, accurate record, accurate deduction from data, all of which may be reduced to simple, plain honesty. Anything else is error, not science. It is not that “honesty is the best policy,” but that in science honesty is the only possible policy. Hence, scientific training should give to every student this one at least of the cardinal virtues, and we may claim with justice this advantage as one of the results to be derived from pursuing scientific studies. In fact the relation of science and biological science, no less than any other, to general schemes of education, has been one of its most important contributions to humanity.

Biology has influenced modern education both in the matter taught and the method of its presentation. It has gone farther and farther into the mysteries of nature and opened up wider fields of knowledge. It has insisted that the student should be trained not only in the facts and the accurate interpretation of facts, but in the methods by which facts may be obtained, thus providing for the continuous growth of the substance from which its principles may be verified and definite conclusions reached.

In recent years there has been a wide demand for the more general distribution of knowledge of nature, and “nature study” has had a prominent place in the discussions of educators. I must confess to some fear for the

outcome of well meant efforts to crowd such studies into the hands of unprepared teachers, though surely no one could wish more heartily for a wider extension of such work well done. It is encouraging to note steady progress in this line and we should be content not to push ahead faster than conditions will warrant.

Our science is an evergrowing one, and I wish to mention briefly some of the conditions of biological research and the conditions essential to its successful prosecution. The time has passed when it is possible for the isolated individual to accomplish much of anything of value in the growth of science. Such instances as the cobbler naturalist can not well be repeated under present conditions, and biological workers must expect that some part at least of their time is spent where libraries, museums and scientific workers are to be found. I recall meeting some years ago in an obscure little village, with a young man who was following a trade, but whose ardent love for nature had brought him to take up the study of a certain group of insects, and in this group he had conceived the idea of preparing a work covering the geographical distribution for the world. With scarcely the beginning of a library, with no access to general collections, apparently with no conception of the stupendous nature of the task he was so ambitiously undertaking, there was perhaps little danger of his discovering the hopelessness of his case. He doubtless gained much pleasure and individual profit in the quest, but for the progress of science, how futile such attempts. Isolated work is often necessary, often the only way in which certain data can be secured, but if isolation be permanent, if it means to be cut off from the records of what has already been done in one's line of study, progress is painfully slow and results of little value. Access then to the world's storehouses of knowledge, to libraries and museums where one may determine the conditions of progress on any given problem is an imperative condition to satisfactory research.

Another condition almost as imperative is time for extended and consecutive work. There are comparatively few places where, after passing the stages of preparation, one may have the opportunity to give uninterrupted time to pure research, but fortunately such opportunities are increasing.

Another factor is necessary equipment, a condition varying indefinitely with the problem undertaken. Studies of some of the simpler processes of

life may be successfully carried on with barely any apparatus whatever, while others require the most costly and complex of machinery. Deep sea investigations, for example, are possible only with a suitable vessel and elaborate apparatus for dredging and other operations, and such expeditions as that of the Challenger, the Blake, the Albatross and others involve such vast outlays that only the liberality of nations or of the very wealthy render them possible.

However, the modest student without a dollar to invest in these expensive undertakings may have the opportunity to work as diligently and effectively as any. So, too, the costly equipments of marine stations, of universities, of national and state museums are open to every earnest worker.

Still another condition related to the best effort in research is a satisfactory outlet for publication. Probably no investigator enters on an elaborate extended research without the expectation that such results as he may obtain, especially such as are novel and important to the growth of science, shall at some time be given a public hearing and a permanent record in the annals of science. However much this ambition may be overworked and abused, it must be considered the logical and legitimate outcome of research, valuable as an incentive to work, essential to the progress of science.

The output of scientific laboratories is always pressing hard upon the organs of publication, and though we have numerous periodicals open to all, many society proceedings and transactions devoted to their membership, university bulletins intended primarily for the staff and students of each institution, still adequate publication facilities are often wanting. Especially is this true regarding the suitable illustration of papers which depend largely on plates or drawings for the elucidation of the text. Our own modest effort in *THE NATURALIST* is an attempt to meet one phase of this demand, but you all appreciate, I think, that it is insufficient for the needs of our own institution. Some of the more extended papers resulting from the work of either students or faculty must suffer oblivion, delay or inadequate presentation. Evidently a publication fund is one of our pressing needs.

Opportunities for research have been much increased within recent years, and now it is possible for one to look forward with some assurance to a career in research pure and simple if that is his desire. As many of those

present doubtless anticipate such career, it may not be amiss to mention some of the opportunities that now present. Positions in connection with universities and colleges now as for a long time past offer some of the most available openings. Fellowships, and positions as assistants with comparatively light duties with expectation that the holder will devote himself to investigation that will advance his branch of science are offered in many places and their value is shown by the numerous candidates for each position. Many government positions in Department of Agriculture, Geological Survey and Fish Commission demand a high degree of training and offer exceptional opportunities for research.

The first few years following graduation are golden days of opportunity in the way of research. For the majority, perhaps, these are the days when the greatest amount of original study may be possible and under conditions favoring the greatest productivity. As time passes and duties and responsibilities increase the opportunity for uninterrupted work grows less and less. Of course original work should follow necessary preparation but can not be postponed indefinitely, in hopes of a more favorable season, if the individual hopes to accomplish anything of value in his chosen science. Too early publication however is to be discouraged. Most good things will keep for a time at least, and the opportunity to test and verify investigations before publishing is desirable. It is unwise to attempt to harvest a crop of glory, in scientific fields at least, before the seed has had time to germinate. The extremes of too hasty publication and indefinite delay are both to be avoided.

But this disjointed address must be brought to a close, I have indulged in a medley rather than pursuing a connected theme, but it has been in my mind to show how the influence of modern biology has been felt in every phase of human life and modified every phase of human thought. It touches history and illumines it as a record of human activities, the modifications and adaptations of the most dominant organism of earth. It touches language and infuses it with life as the highest evolution of all means of communication among animals. It enters the sphere of human relations and we see society, government, law, as the most complex expression of forces operative all along the line of organic life.

We may gain inspiration in our work from the thought that our field of labor gives opportunity for the highest service in the advancement of human

interests and the intellectual uplift of the race.

The club extended Prof. Osborn a vote of thanks for his valuable address.

F. J. TYLER, *Secretary*.

The Ohio State Academy of Science held its eleventh annual meeting at the Ohio State University in this city on November 29th and 30th. Between thirty and forty papers were given and the attendance was considerably above the average. On the evening of the 29th a joint meeting was held with the Modern Language Association of Ohio, which held its annual session at the University on the dates mentioned above. The committee arranged an interesting and appropriate program for the evening and a large and appreciative audience responded. The Academy meetings have been held heretofore during the Christmas vacation, therefore holding it at this time was an experiment, but judging from the program, attendance, and enthusiasm manifested, the meeting this year may be said to be one of the best the society has ever held.

It is of more than ordinary interest to be able to record the taking of specimens of the European ruff, *Pavoncella pugnax* (Linn.) in Ohio. Two male specimens are in the Dr. Jasper collection at the Ohio State University, one taken April 28th, 1879, at Columbus, the other November 10th, 1872, at the Licking Reservoir.

FIFTY ADDITIONS TO THE CATALOGUE OF OHIO PLANTS.

W. A. KELLERMAN.

The plants listed below have been found growing in the State without cultivation. A large number of them are adventive species but not hitherto recorded in the Ohio list. Three of the names occurred in the old lists and were noted in the Catalogue of 1893 by Kellerman and Werner, but were discarded in the Fourth State Catalogue, published in 1899. These here referred to and which are below restored to the Ohio list, are Nos. 683*a*, 1423*b*, and 1990½*a*. No. 893*a* was included in L. D. Stair's list of Railway Weeds. All the others are wholly new to the listed flora. While several persons have contributed to this increase, special thanks are due to Mr. Otto Hacker, who formerly as well as at present, contributed largely to a fuller knowledge of the State flora. Mr. Hacker has furnished specimens of all the species credited to him below and these are deposited in the State Herbarium. The rich field for adventive species in the region of Painesville may be understood when it is stated that the extensive and long-established nursery grounds of Storrs and Harrison are located at this place.

- 1*a*. *Botrychium lunaria* (L.) Sw. Moonwort. Painesville, Lake Co. Otto Hacker.
- 201*a*. *Apera spica-venti* (L.) Beauv. Silky Bent-grass. Wild-straw. Painesville, Lake Co. Otto Hacker.
- 201*b*. *Aira caryophyllea* L. Silvery Hair-grass. Rarely escaped. Painesville, Lake Co. Otto Hacker.

- 253a. *Festuca myuros* L. Rat's-tail Fescue-grass. Painesville, Lake Co. Otto Hacker.
- 272a. *Hordeum sativum* Jessen. Common Barley. Occasionally escaped.
- 272b. *Hordeum distichum* L. Two-rowed Barley. Rarely escaped. Painesville, Lake Co. Otto Hacker.
- 470a. *Lemna cyclostasa* (Ell.) Chev. (*L. valdiviana* Phil.) Valdivia Duckweed. Richmond, Lake Co. Otto Hacker.
- 557a. *Gemmingia chinensis* (L.) Kuntze. Blackberry Lily. Escaped. Franklin Co. J. H. Schaffner.
- 557b. *Crocus vernus* All. Crocus. Escaped. Painesville, Lake Co. Otto Hacker.
- 568a. *Limnorchis hyperborea* (L.) Rybd. (*Habenaria hyperborea* (L.) R. Br.) Canton. Mrs. Theano W. Case.
- 670a. *Quercus alexanderi* Britton. Alexander's Oak. "Ohio;" N. L. Britton, Manual of Flora, 336. This was formerly confused with, or included in *Q. acuminata*, and like the latter is not uncommon in Ohio.
- 683a. *Urtica urens* L. Small Nettle. Painesville, Lake Co. Otto Hacker.
- 754a. *Acnida tamariscina prostrata*. Uline and Bray. Painesville, Lake Co. Otto Hacker.
- 762a. *Portulaca grandiflora* Hook. Garden Portulaca. Sun Plant. Escaped; Roadsides. St. Marys, Auglaize Co. A. Wetzstein.
- 775a. *Lychnis vesicaria* L. Lychnis. Escaped. Painesville, Lake Co. Otto Hacker.
- 886a. *Fumaria parviflora* Lam. Small Fumitory. Painesville, Lake Co. Otto Hacker.
- 893a. *Sisymbrium altissimum* L. Tall Sisymbrium. L. D. Stair in List of Railroad Weeds. Painesville, Lake Co. Otto Hacker.
- 894a. *Myagrum perfoliatum* L. Myagrum. Painesville, Lake Co. Otto Hacker.

- 921a. *Camelina microcarpa* Andrz. Small-fruited False-flax. Painesville, Lake Co. Otto Hacker.
- 984a. *Rubus neglectus* Peck. Purple Wild Raspberry. Painesville, Lake Co. Otto Hacker.
- 985a. *Rubus phoenicolasius* Maxim. Japan Wineberry. Escaped from cultivation; comes freely from seed, and propagates by tips. Painesville, Lake Co. Otto Hacker.
- 1004a. *Potentilla pumila* Poir. Dwarf Five-finger. Painesville, Lake Co. Otto Hacker.
- 1026b. *Sorbus aucuparia* L. European Mountain Ash. Escaped. Painesville, Lake Co. Otto Hacker.
- 1051a. *Prunus mahaleb* L. Mahaleb. Perfumed Cherry. Columbus, Franklin Co. W. A. Kellarman. Painesville, Lake Co. Otto Hacker.
- 1054a. *Acuan illinoensis* (Mx.) Kuntze. (*Desmanthus brachylobus* Benth.) Illinois Mimosa. New Richmond, Clermont Co. A. D. Selby.
- 1071a. *Trifolium dubium* Sibth. Least Hop-Clover. Painesville, Lake Co. Otto Hacker.
- 1091a. *Coronilla varia* L. Coronilla, Axseed, Axwort. Painesville, Lake Co. Otto Hacker.
- 1122a. *Vicia augustifolia* Roth. Smaller Common Vetch. Painesville, Lake Co. Otto Hacker.
- 1171a. *Euphorbia cuphosperma* (Englem.) Boiss. Warty Spurge. Painesville, Lake Co. Otto Hacker.
- 1195a. *Euonymus europaeus* L. Spindle-tree. Escaped. Painesville, Lake Co. Otto Hacker.
- 1265a. *Viola odorata* L. English or Sweet Violet. Escaped. Painesville, Lake Co. Otto Hacker.
- 1301b. *Kneiffia linearis* (Mx.) Spach. Narrow-leaf Sun-drops. Painesville, Lake Co. Otto Hacker.

- 1423b. *Spigelia marylandica* L. Indian Pink or Carolina Pink. Fl. M. V. A. P. Morgan. North Madison, Lake Co. D. W. Talcott.
- 1502a. *Asperugo procumbens* L. German Madwort. Catchweed. Painesville, Lake Co. Otto Hacker.
- 1534a. *Scutellaria parvula ambigua* Fernald. "Ohio," Nuttall. Greene Co., E. L. Moseley; Montgomery Co., W. U. Young; Franklin Co., E. E. Bogue; Gallia Co., J. W. Davis.
- 1556a. *Salvia lanceolata* Willd. Lance-leaf Sage. By roadside near Columbus. W. A. Kellerman.
- 1586a. *Mentha longifolia* (L.) Huds. Horse Mint. Painesville, Lake Co. Otto Hacker.
- 1600a. *Physalis francheti* Mast. Chinese Lantern Plant. Escaped. Painesville, Lake Co. D. W. Talcott.
- 1609½a. *Datura metel* L. Entire-leaf Thorn-apple. Escaped. Painesville, Lake Co. Otto Hacker.
- 1611b. *Kickxia spuria* (L.) Dumort. (*Elatinoides spuria* Wetzst.) Painesville, Lake Co. Otto Hacker.
- 1690a. *Diodia teres* Walt. Rough Button-weed. Painesville, Lake Co. Otto Hacker.
- 1702a. *Asperula hexaphylla* All. *Asperula*. Escaped. Painesville, Lake Co. Otto Hacker.
- 1712a. *Viburnum lantana* L. Wayfaring Tree. Escaped. Painesville, Lake Co. Otto Hacker.
- 1732a. *Valeriana officinalis* L. Garden Valerian. Escaped. Painesville, Lake Co. Otto Hacker.
- 1756a. *Arnoseris minima* (L.) Dumort. Lamb Succory. Painesville, Lake Co. Otto Hacker.
- 1756b. *Hypochaeris glabra* L. Smooth Cat's-ear. Painesville, Lake Co. Otto Hacker.
- 1766c. *Lactuca virosa* L. Strong-scented Lettuce. Confused with *L. scariola* according to Britton, being the commoner of the two

species. (A. D. Selby, Meeting Ohio Academy of Science,
November, 1901.)

1775*a*. *Hieracium pilosella* L. Mouse-ear Hawkweed. Painesville, Lake
Co. Otto Hacker.

1990½*a*. *Tanacetum vulgare crispum* DC. Tansy. Painesville, Lake Co.
Otto Hacker.

BOTANICAL CORRESPONDENCE, NOTES AND NEWS FOR AMATEURS, I.

Conducted by W. A. KELLERMAN.

Item 1. It has been asked how many species of plants occur in Ohio. Only a guess can at present be made. In the Catalogue of Ohio Plants, by Kellerman and Werner, prepared in 1893, there were listed 1,925 Spermatophytes, 68 Pteridophytes, 335 Bryophytes, and 1,400 Thallephytes. The Fourth Catalogue, by the writer, published in 1899, gave 2,025 species of Pteridophytes and Spermatophytes. While many additions to the previous list were included, very many species formerly reported were excluded because unauthenticated by herbarium specimens, and others were undoubtedly extra-limital for Ohio. Two Annual Supplements to this catalogue have been issued, bringing the number of species of the vascular plants, nearly all authenticated, up to about 2,150. The mosses, the higher fungi and the lichens have been listed with some degree of fullness, but most of the other lower plants have been very incompletely placed on record, though large collections, only partially worked up as yet, are now in the herbarium of the State University.

Item 2. Miss Ruth E. Brockett, of Rio Grande, Gallia County, Ohio, has found the Showy Skullcap, *Scutellaria serrata* Andr., previously unreported for this State. The distribution, as given in Britton's Flora, is New York and Pennsylvania to North Carolina, Illinois and Kentucky. In the Rio Grande region many interesting or new plants for the Ohio list have hitherto been detected by Miss Brockett, as the Fringe Tree (*Chionanthus virginica*), the Purplish Buckeye (*Aesculus octandra hybrida*), and others too numerous to mention.

Item 3. An interesting and suggestive study has been published by Herman Dingler (Muenchen) on the organs for wind-dispersal (flug-organe) in the Vegetable Kingdom. The title of the book is “Ein Beitrag zur Physiologie der passiven Bewegungen im Pflanzenreich.” After describing fully the mechanics involved, and the methods of investigation, the author enumerates the *Chief Types* of the flight organs as follows (prefixing to the word “flyer” the descriptive words, 1, dust; 2, granule; 3, bubble; 4, hair; 5, pan; 6, umbrella; 7, sail; 8, disk-twist; 9, barrel-twist; 10, plain-twist; 11, screw, and 12, screw-twist):

I. Group.

1. Staubflieger, e. g. Micrococcus, Puffball, Spores of Mosses, Pollen.
2. Körnchenflieger, e. g. Poppy, Species of the Pink Family, Orobanchaceæ.
3. Blasenflieger, e. g. Ironwood (Ostrya), Valerianella, Rhus cotinus.
4. Haarflieger, e. g. many Bromeliaceæ, Pitcairnia, etc.
5. Napfflieger, e. g. Wafer Ash (Ptelea trifoliata), Cochleospermum.
6. Schirmflieger, e. g. the Compositæ, Milkweeds (Asclepiadaceæ), Willows.

II. Group.

7. Segelflieger, e. g. Cross-vine and seeds of other Bignoniaceæ.

III. Group.

8. Scheibendrehflieger, e. g. flattened seeds of the Iris, and Lily families.
9. Walzendrehflieger, e. g. Silver bell (Halesia), Knotweed (Polygonum), etc.
10. Plattendrehflieger, e. g. Tecoma stans. (The Ailanthus type.)

IV. Group.

11. Schraubenflieger, e. g. Maples, genera of Coniferæ, Sapindaceæ, etc.

V. Group.

12. Schraubendrehflieger, e. g. fruit of Liriodendron tulipifera.

Item 4. The recent death of Thomas Meehan, horticulturist and botanist, removes from the list of active American workers one whose numerous, accurate and original observations contributed greatly to the advancement of botanical science.

NOTE AND CORRECTION TO OHIO FUNGI EXSICCATI.

W. A. KELLERMAN.

A critical inspection of the nomenclature used for the first Fascicle of the Ohio Fungi might seem to warrant the conclusion that the judgment of more recent workers is sometimes ignored and that a too conservative course has been adopted. But it should be remembered that the main purpose is to furnish Ohio material accompanied by names (occasionally synonyms) that were undoubtedly applied to the species represented. I have preferred to use for the Rust on Sunflower, *Puccinia helianthi*, rather than *P. tanacetii*—recent work on other species suggesting that with this also when fully studied, a physiological distinction may supplement the too insignificant morphological difference. Again, I have used *Aecidium album*, which Clinton applied to the first stage of the Uredine found on *Vicia*, not ignorant of the fact that Dietel gives this as a stage of *Uromyces albus*—but should not this first be substantiated by cultures? It is to be added that through inadvertency Peck's later name (*Aecidium porosum*) was used, hence here follows a corrected label with both Clinton's and Peck's descriptions:

2. Aecidium album Clinton.

Aecidium porosum Peck.

On *Vicia americana* Muhl.

Lakeside, Ottawa Co., O. May 17, 1901.

Coll. W. A. Kellerman.

“*Aecidium album* Clinton, spots none; peridia scattered, short, white, the margin subentire; spots subglobose, white, about .0008 inches in diameter.” Report on the State Museum, State of New York, 26:78. 1873.

“*Aecidium porosum*, Pk. Spots none; cups crowded, deep-seated, broad, wide-mouthed, occupying the whole lower surface of the leaf to which they give a porous appearance; spores orange-colored, subangular, .0008–.001 inch in length.” Botanical Gazette, 3:34. April, 1878.

NOTES OF TRAVEL IN PORTO RICO.

ROBERT F. GRIGGS.

By its configuration, Porto Rico is divided into two parts very distinct from each other in almost every respect and of primary importance in all the affairs of the island. The north side, which comprises about two-thirds of the total area, is kept constantly wet with almost daily rains. On the south it has been known not to rain for a whole year in some places. On the north side grows an abundance of luxuriant, tropical vegetation; on the south in many localities are barren hills covered only with scrub brush. But throughout the island there is great local variation in all the climatic and physical conditions.

Along most of the north side there stretches a low, coast plain, out of which rise numberless, small, steep hills. This plain, everywhere well watered, is in most places very fertile, but in the vicinity of Vega Baja it becomes a sandy waste. This sand desert is one of the most peculiar places it has ever been my fortune to visit. There is no grass (turf-making grass is almost unknown in the tropics), neither are there large trees. Everywhere are low bushes not much more than ten feet tall. The sand beneath them is bare in many places, but is covered in others with various forms of herbage, most of which, instead of being composed of desert forms, as would be expected, is made up of the most typical water-loving plants, among which, *Sphagnum* (two species) and *Utricularia* are noteworthy. Imagine, if you can, a sphagnum bog shading into loose sand in a distance of only ten feet with no change in level. The explanation of this peculiar fact is, however, not hard to find. The rainfall is so copious that wherever there is any means of holding it, the hydrophytes take hold and spread, themselves acting as

water holders when once started, while in other places the water quickly soaks into the sand and leaves it as dry as ever.

The plain on which this sand desert is located is separated in most places from the sea by low hills. It is very level and was probably once covered with water out of which projected many rocky islands—the limestone hills of to-day. These hills are a very characteristic feature of the country. From an incoming vessel they are plainly seen projecting like saw teeth all along the coast; from an eminence back in the country they appear to have no system or regularity whatever, but stick up anywhere sharp and rugged as though shaken out of a dice box onto a board. Further inland they are closer together with no plain between, though in other respects like those of the coast. It is as though they were eroded when the sea stood lower than it does to-day, perhaps very much lower; then the valleys were filled up during a period when the sea was slightly higher than at present, whence it has receded and left the island of to-day. They are covered with a characteristic jungle, rising conspicuously out of which is the “Llume” palm (*Acria attenuata*) whose graceful stem, only about half a foot thick at the base, attains a height of a hundred feet, tapering till it is only three or four inches thick at the top. It is nearly white and at a distance entirely invisible, so that the crown of leaves looks as though it were floating around in the air above the surrounding vegetation.

Further inland the limestone hills give way to others of red clay. The clay, like the limestone, is very deeply eroded. In most places it is so continually washed down that the sides of the hills stand always at the critical angle and are ready to slide from under the feet of the explorer. Indeed it would be impossible to climb them were it not for the numerous bushes everywhere standing ready to lay hold on. Here abound ferns, Melastomaceae and other plants of humid regions. Tree ferns are very common; the largest belong to one species of *Cyathia*. Its beauty is simply beyond description. Imagine, you who have never seen it, a trunk thirty feet tall surmounted by a crown of a dozen or fifteen great leaves made up of a score or two pinnae of the size and grace of ordinary ferns and you have the components—not the ensemble—of the tree fern.

This red clay region is the land of coffee. Everywhere the novice thinks the hillside covered with jungle, which turns out to be only poorly kept coffee plantations. The coffee region is coextensive with the range of

several plants. Two or three species of the pepper family, with large peltate or round leaves, are found only here; and with one or two exceptions the Melastomaceae occur only in this wet country. They are a very large group of plants common throughout the tropics, but represented in the northern states by the common *Rhexia*. Its members may be known anywhere by their three-nerved leaves, many of which are beautifully patterned and marked so that even among other tropical plants they are conspicuous for their beauty.

When we cross the summit we come upon a different sort of vegetation; cacti take the place of tree ferns, and instead of wet jungles we have dry scrub brush full of spiny and thorny shrubs with almost every sort of prickly one can think of. One who has never encountered them can scarcely appreciate the abundance and effectiveness of tropical thorns. These thickets of brush extend over most of the undisturbed portion of the south side. Everywhere through them there are scattered cacti of several sorts; but near Guayanilla, a few miles west of Ponce, these become relatively much more numerous so as to form a veritable cactus desert. Only here is the largest form present. It is a large *Opuntia* with a bare stem and long arms radiating in one or two whorls near the top. Besides it there are several species of *Cereus* and another small *Opuntia* similar to the common prickly pear, together with a species of the same group cultivated for its fleshy branches which are eaten. All through this dry region agaves or century plants are very common. There seem to be several species, but they are such terrors to botanists that it is hard to tell anything about them.

From this brief sketch it will be seen what a diversified flora Porto Rico offers to the student. There are opportunities for several ecological studies of surpassing interest, and on the systematic side the work has only been begun. At present there are scant facilities for the student, but with the fuller occupation of the island by American government and customs, we may hope that some of our enterprising universities will establish there a school of tropical agriculture and botany, fields now white for the harvest but almost without workers.

Washington, D. C., October 30, 1901.

SALAMANDERS TAKEN AT SUGAR GROVE.

MAX MORSE.

On May 25, 1901, Prof. Hine, while collecting in the hills at Sugar Grove, Fairfield County, O., found a salamander under a piece of pine log on the slope of a hill, about a hundred yards from water. It was, for the time, put in a jar along with several individuals of *Desmognathus fusca* Raf., which were taken in, or within a few feet of the rivulets which flow down the valley. Aside from this specimen taken on the hillside, all the specimens were found not farther than a half dozen feet from the water. When the collections were examined in the laboratory it was found that the single specimen just mentioned differed in many respects from the others. This led to investigation and it was found that it corresponded closely with the description of *D. ochrophæa* Cope. Thus, the posterior portion of the mandible was edentulous; no tubercle in canthus ocelli; belly paler than in any of *D. fusca* taken; length nearly three-fourths of an inch shorter than the others; a light bar from eye to corner of mouth; tongue free behind; parasphenoid teeth separated behind. The specimen was kindly examined by Dr. J. Lindahl, of the Cincinnati Society of Nat. Hist., who is acquainted with the form. He agreed that it corresponded with the description of Cope. Whether the characters as given above are sufficient to place the specimen under *ochrophæa* is a matter hard to decide. Cope gives the range of *ochrophæa* as "in the Alleghenies and their outlying spurs." Dr. Lindahl has a specimen from Logansport, Ind., taken November 10, 1900.

FISHES TAKEN NEAR SALEM, OHIO.

E. B. WILLIAMSON.

The present short list is published, not because of any records of special interest, but in order that a record may be made of the fish known certainly from the headwaters of Beaver Creek. In the case of fish the most logical and significant way to indicate distribution is certainly by streams, and a very small contribution to the ichthyology of the above named stream is here presented.

About three-fifths of Columbiana County is drained by Beaver Creek, one-fifth by the Mahoning River and streams leaving the county to the west, while the remainder enters the Big Yellow and Little Yellow Creeks. Beaver Creek is practically confined to Columbiana County, though it empties into the Ohio River in Pennsylvania at Smith's Ferry, just above the state line. The relation of Beaver Creek to the Mahoning River is interesting, the two being in general, arcs of concentric circles with the Mahoning outside. A person going directly west from Salem crosses Middle Fork of Beaver Creek first, then the Mahoning, and the same is true if he goes directly north or directly east. South-west of Salem the small streams emptying into the Mahoning have not been seined. From one of these Herman McCane has taken a specimen of *Ichthyomyzon concolor* which is preserved in the Salem High School collection with the other species here recorded. All the other streams in close proximity to Salem are part of the system of the Middle Fork of Beaver Creek, with the exception of Cold Run, which flows almost directly south into the West Fork of Beaver Creek, the stream thus formed soon being augmented by the waters of the North Fork.

Seining has been done only near Salem in small tributaries and where Middle Fork has an average width of not more than ten or twelve feet. Mr.

Albert Hayes, Mr. J. S. Johnson and Mr. F. W. Webster have helped me draw the seine. Mr. Webster has also given me many valuable suggestions as to suitable localities.

1. *Ameiurus melas* (Raf.). Rare, only in main stream.
2. *Catostomus commersonii* (Lac.). Common, main stream and tributaries.
3. *Catostomus nigricans* Le S. Taken only in a small tributary.
4. *Moxostoma aureolum* (Le S.). In a small tributary.
5. *Cyprinus carpio* L. Only in main stream.
6. *Campostoma anomalum* (Raf.). Everywhere.
7. *Chrosomus erythrogaster* Raf. In two small tributaries.
8. *Pimephales promelas* Raf. In main stream only.
9. *Pimephales notatus* (Raf.). Everywhere.
10. *Semotilus atromaculatus* (Mitch.). Everywhere.
11. *Leuciscus elongatus* (Kirt.). In one tributary and in Cold Run. The iridescent coppery red of the sides anteriorly in the living fish, taken in October, turned scarlet in alcohol.
12. *Abramis chrysoleucas* (Mitch.). Taken only in main stream.
13. *Notropis cayuga* Meek. A single specimen taken in Cold Run.
14. *Notropis cornutus* (Mitch.). Everywhere.
15. *Notropis rubrifrons* (Cope). Taken only in main stream.
16. *Notropis umbratilis lythrurus* (Jordan). Taken only in main stream.
17. *Ericymba buccata* (Cope). Everywhere.
18. *Rhinichthys atronasmus* (Mitch.). In the smallest tributaries.
19. *Hybopsis amblops* (Raf.). In Cold Run.
20. *Hybopsis kentuckiensis* (Raf.). Taken only in Cold Run, a single specimen.
21. *Umbra lima* (Kirt.). Taken only in the main stream.

22. *Eucalia inconstans* (Kirt.). Taken only in the main stream.
23. *Ambloplites rupestris* (Raf.). In main stream and one tributary.
24. *Apomotis cyanellus* (Raf.). Taken in Cold Run.
25. *Lepomis megalotis* (Raf.). Taken only in the main stream.
26. *Eupomotis gibbosus* (Lin.). One specimen taken in a tributary; determined by Dr. Evermann. Numbers 25 and 26 probably represent one species.
27. *Micropterus dolomieu* Lac. Taken only in the main stream.
28. *Boleosoma nigrum* (Raf.). Everywhere.
29. *Etheostoma flabellare* Raf. In the main stream and Cold Run.
30. *Cottus ictalops* (Raf.). Taken only in Cold Run.

MR. A. J. PIETERS, Assistant Botanist in the U. S. Dept. of Agriculture, has written an interesting and useful article^[1] on the plants of western Lake Erie. This report should be read by all who are interested in the hydrophytes of Ohio, or in the flora and fauna of Lake Erie. In addition to some introductory remarks, the paper treats of the plants in Put-in-Bay, in Squaw Harbor, near Gibraltar Island, in Hatchery Bay and in the open lake, and the plants of East Harbor. The swamp vegetation is also discussed, including the plants in the Portage River swamps and in the swamps about Sandusky Bay. The ecological conditions and the ecological adaptations of the flora are treated quite fully, and at the end are given alphabetical lists of the plants studied, including angiosperms, stoneworts and desmids.

JOHN H. SCHAFFNER.

¹. A. J. PIETERS. "The Plants of Western Lake Erie, with Observations on their Distribution." Bull. U. S. Fish Commission, 1901, pp. 57-79. Pls. 11-20.

COLLECTING TABANIDÆ.

JAMES S. HINE.

The habits of flies belonging to the family Tabanidæ, commonly called horse-flies or gad-flies, furnish much material for study and observation. I take this opportunity to record some of the notes which I have taken in the last few years while endeavoring to collect and study the local species of the family. Although the eggs, larvæ and pupæ of many species have been studied, what I have to say in this paper pertains wholly to the adults. Members of the family are usually taken by every entomologist who does general collecting, but as a usual thing males are seldom taken; in fact this sex is so poorly represented in collections that no key has been published for identifying the males of our American species. The student must use the key to the females as far as possible and guess at the rest. In very many cases the male is not even described, so that sometimes, when the sexes are unlike, they can be associated only by observations in the field. By careful collecting and observation we have procured practically all of our local species in both sexes, and the derived benefit, satisfaction and enjoyment have paid us fully for our time and pains.

In the first place the mouthparts of the two sexes are different—the male lacks the mandibles which are present in the female. This makes it necessary for them to procure their food from different sources, the male obtains his from flowers, while the female lives by puncturing the skin and sucking the blood of warm-blooded vertebrates. Thus it is evident that during the time spent in procuring food the sexes cannot remain together. From an economic standpoint the female most concerns the student and she is often taken for study without an attempt being made to procure the male.

At this point I can say collect females around horses, cattle and other animals, and males on flowers; but this is not enough, for knowing the general habits of insects we are certain that there is a common ground where the two sexes may be found together. One finds this common ground in the vicinity of water, where their transformations take place and where their eggs are laid, also in various other places, which we shall take occasion to discuss as we proceed.

The females of all our local species of *Chrysops* with *Tabanus pumilus* and *nivosus* come buzzing around the collector in numbers, and at such times may be taken easily with a net. Other species of *Tabanus* come near enough that the sound of their wings is recognizable, but are so active that it is almost impossible to procure them.

During the time the female is ovipositing the male is often sitting near by on the foliage. At Georgesville, Ohio, June 4th, I observed *C. mœchus* ovipositing on foliage overhanging a mill-race; soon after specimens of the male sex were observed resting on the upper leaves of the same plant on which females were ovipositing. In a few minutes collecting, a dozen or more specimens of each of the sexes were procured. The only males of *C. indus* I have ever taken were procured at Columbus, on the border of a small pond, where the females were ovipositing.

The sexes of many species of *Tabanus* often alight on the bare ground of paths or roads that run through or along woods. At Cincinnati, June 10th, in company with Mr. Dury, we procured large numbers of the sexes of different species resting on some furrows that were plowed around a woods to prevent the spread of fire. We also took the same species resting in paths and roads that ran through the woods. Some of these same species were also taken from low-growing foliage in sunny places among the trees. At Medina, Ohio, males and females of *T. vivax* and *trimaculatus* were taken while resting in a road that ran through a dense woods.

One of the best places I have ever found to get the sexes of *Chrysops* and *Tabanus* is in the tall grass that skirts the marshes of Sandusky Bay. This grass is the *Phragmites* of botanists and grows to a great height by July 1st. On July 6th, at Black Channel, when the wind was high I went into a patch of this grass that was so dense that I could not use a net to advantage. Here I saw an abundance of flies and found that by approaching them very slowly I could readily pick the specimens off with my fingers. The male and

female of *T. stygius*, *nivosus*, *C. æstuans* and *flavidus* and the male of *T. affinis* and *bicolor* were taken in this way. I found that this same species of grass afforded excellent collecting wherever found, but most material was procured when the wind was high. On the same date and near the same place the male of *C. flavidus* was taken from the flowers of the common spatter-dock, and this and *æstuans* were procured by sweeping in the adjacent low-growing herbage. R. C. Osburn informs me that he has had excellent success in collecting Tabanids from tall grass near water in his experience.

Tabanus sulcifrons Macq. is an abundant species in northern Ohio during the latter part of July and all of August. So common that by actual count twenty-eight specimens were taken from a cow in ten minutes, while a few that alighted on the animal during that time were not procured. August 1st of the present year I was at Hinckley, Medina County, and spent the day taking observations on this species. In the morning about nine o'clock I went to the border of a woods where I had often observed the species before. Here males and females were found in abundance crawling over the trunks and foliage of trees, on the fence along the woods and flying about generally. One pair was observed in copulation on the fence, and I am of the opinion that the presence of so many flies in the locality at the time is explained on the ground that it was the general mating place of the sexes. On several occasions I have made observations which lead me to believe that the sexes of various species of the family copulate among foliage often high up in the trees. As Tabanids are not easily procured with a net from the surface of a rough rail, I tried the experiment of picking the specimens off with my fingers and found that it was surprisingly successful, if the movement toward them was made very slowly until just ready to touch them when the fingers were gripped quickly. Near a watering trough where a herd of cattle drank daily I found males in numbers resting on the ground where the turf had been tramped off. Along Rocky River I observed both sexes fly down to the water and dip several times in succession and then away to alight on a stone on the bank or disappear from sight altogether.

On July 29th I rode from Sandusky to Cleveland by boat. Although we were from two to five miles off shore all the time, males and females of *T. sulcifrons* often came on board and alighted on the canvas and rigging of the boat. From this it is evident that this species at least may fly for some distance over water.

We have taken *Goniops chrysocoma* on several occasions. It has a habit which is of value to the collector. At Hinckley, Medina County, I took several females and observed that they have the habit of stationing themselves on the upper side of a leaf, where by vibrating their wings rapidly and striking the upper surface of the leaf at each downward stroke, make a rattling noise which can be heard plainly several feet away. At Vinton last spring Mr. Morse and myself identified the characteristic sound of the species and were guided by it to procure specimens.

I have taken the male of *Pangonia rasa* on blossoms of sumac at Medina, Ohio, in August.

OBSERVATIONS ON INSECTS.

JAMES S. HINE.

AGROMYZA SETOSA Loew—The larvæ of several species of the genus *Agromyza* are known to mine the leaves and stems of various plants. Cabbage, potatoes, corn, clover, strawberries, verbenas, chrysanthemums and sunflowers are among the cultivated plants from which various species of the genus have been reared; while plantain, round-leaved mallow, golden-rod, aster, cocklebur, rag-weed and wild-rice are given as their food-plants. In some cases a single species of fly has been reared from a half dozen or more different plants. *Agromyza setosa* Loew, as determined by Coquillett, was reared in numbers from leaves of wild-rice, *Zizania aquatica*, at Sandusky during August of each of the years 1900 and 1901. Professor Osborn studied the species and its work in 1900, while my observations were made a year later. Although I include the notes taken by both of us, many points are needed before a detailed account of the habits and life history of the species can be given.

The eggs are conspicuous on account of their abundance and white color, and are deposited chiefly on the upper surface of the leaves of the food plant.

The larvæ upon hatching bore into the leaf and feed beneath its upper covering. When full grown they measure about 6 mm. in length, are white, or greenish on account of chlorophyl taken in with their food, and are furnished with strongly chitinous mouth parts. The mines which they make in the leaves are irregular in width and extend for varying lengths on one side or the other of the mid-rib. These variations in extent are usually explainable from the fact that a variable number of larvæ occupy the different mines. The work of the larvæ is apparent from the first on the

upper side of the leaf, and may be seen beneath after a few days because of the fact that the parts beneath the mine sooner or later turn yellow.

The pupa is to be found either in the mine or clinging to the surface of the leaf. It is brown in color, with two prominences anteriorly where the attachment with the leaf is effected, and is contained within the last larval skin so that the legs and wing-pads are at no time visible from the outside.

BIBIO ALBIPENNIS Say—Larvæ observed in colonies under fallen logs, and boards which were lying on the ground. Specimens taken April 4th pupated May 5th and the adults appeared May 13th. The adults were unable to fly for several hours after they emerged on account of their wings remaining soft. I observed the first males flying out of doors on the 23d of May.

CHRYSOPIA ORNATA Say—Larva about an inch and a half in length, white in color, cylindrical, with an enlargement at the posterior end bearing a number of fleshy elongations which are about the length of their basal breadth. Found under rotten wood May 1st. Pupa brown, last segment armed with six spinose teeth, the two on the ventral side arising from the same base, the remaining abdominal segments furnished with a circlet of spines near the posterior third. The adult emerged the 18th of June.

Ohio State University.

Six Colleges well equipped and prepared to present the best methods in modern education. The advantages are offered to both sexes alike.

The following list of departments will suggest the organization of the institution:

Agriculture, Agricultural Chemistry, American History and Political Science, Anatomy and Physiology, Architecture and Drawing, Astronomy, Botany, Chemistry, Civil Engineering, Clay Working and Ceramics, Domestic Science, Economics and Sociology, Education, Electrical Engineering, English Literature, European History, Geology, Germanic Languages and Literatures, Greek, Horticulture and Forestry, Industrial Arts, Latin, Law, Mathematics, Mechanical Engineering, Metallurgy and Mineralogy, Military Science, Mine Engineering, Pharmacy, Philosophy, Physical Education, Physics, Rhetoric and English Language, Romance Languages, Veterinary Medicine and Zoology and Entomology.

Catalogues will be sent upon application. After examining the catalogue write for specific information to the President

Dr. W. O. THOMPSON, Columbus, Ohio.

SPECIMENS DESIRED FOR

**BOTANICAL
MUSEUM.**

The large mushrooms, Puffballs and other Fungi; Abnormal growths and interesting specimens of shrubs and trees. Also herbarium specimens of Algae, Fungi, Mosses and Ferns as well as flowering plants. Address

Prof. W. A. KELLERMAN,

Department of Botany, Ohio State University,
Columbus, Ohio.

**GEOLOGICAL
MUSEUM.**

Will exchange Hudson, Corniferous and
Carboniferous fossils. Address

Prof. J. A. BOWNOCKER, Curator,
Geological Museum, Ohio State University,
Columbus, Ohio.

**ZOOLOGICAL
MUSEUM.**

Birds, Insects, Reptiles, etc. We wish to make
our collections representative for the fauna of the
state and will greatly appreciate all contributions to that end.

Address, Prof. HERBERT OSBORN,
Department Zoology and Entomology,
Ohio State University, Columbus, Ohio.

Ohio State University, Lake Laboratory. Located at Sandusky on Lake
Erie. Open to Investigators June 15 to September 15. Laboratory
courses of six and eight weeks beginning June 30, 1902. Write for
special circular.

Bucher Engraving Co.

Process and Wood Engraving, Electrotypers and
Manufacturers of Stereotyping and Engraving
Machinery.

80½ North High Street, COLUMBUS, OHIO.

Lepidoptera Price List No. 3.—Price 5 cents Refunded
to Buyers

Issued November 15th, 1901.

Dealers in all kinds of ENTOMOLOGICAL SUPPLIES.

Manufacturers of the Original and Celebrated SCHMITT INSECT
BOXES.

Builders of INSECT CABINETS, Etc.

American Entomological Company
1040 DE KALB AVENUE, BROOKLYN, N. Y.



NEW EDITION



WEBSTER'S
INTERNATIONAL
DICTIONARY

New Plates Throughout.

25,000 ADDITIONAL WORDS

Phrases and Definitions

Prepared under the direct supervision of W. T. HARRIS, Ph.D., LL.D., United States Commissioner of Education, assisted by a large corps of competent specialists and editors.

Rich Bindings. 2364 Pages. 5000 Illustrations.

 *The International was first issued in 1890, succeeding the "Unabridged." The New Edition of the International was issued in Oct., 1900. Get latest and best.*



Also **Webster's Collegiate Dictionary** with
Glossary
of Scottish Words and Phrases.



"First class in quality, second class in size."
NICHOLAS MURRAY BUTLER.

Specimen pages, etc., of both books sent on application.
G. & C. Merriam Co. Springfield, Mass.

SYSTEMATIC COLLECTION OF

Minerals, Rocks, Fossils

Birds and Mammals, Shells and Echinoderms.
Also Human Skeletons, and Anatomical Models.

Relief Models by Messrs. Shaler, Davis and Harris, illustrating

Physical Geology
and Physiography

WRITE FOR CIRCULARS.

Wards' Natural Science Establishment,
ROCHESTER, N. Y.

If you have your PHOTOS made at the **BAKER'S ART GALLERY**,
You will ALWAYS be pleased

State and High Sts., COLUMBUS, OHIO.

OUR LIST OF AWARDS FOR THE LAST YEAR ARE:

The Gold Medal at Paris Exposition.

Highest Award at the Pan-American Exposition.

Six First Premiums out of seven at Ohio State
Exposition.

DIE STAMPING. PLATE AND LETTER PRESS PRINTING.

SPAHR & GLENN,
PRINTERS AND PUBLISHERS.

50 EAST BROAD STREET. COLUMBUS, OHIO.

OHIO MEDICAL UNIVERSITY.

DEPARTMENTS OF

MEDICINE, DENTISTRY AND PHARMACY.

Four years graded course in Medicine, three in Dentistry, and two in
Pharmacy. Annual Sessions, seven months.

All Instruction, except Clinical, by the Recitation Plan.

Students graded on their daily recitations and term examinations. Large
class rooms designed for the recitation system. Laboratories are large, well
lighted and equipped with modern apparatus. Abundant clinical facilities in
both Medical and Dental Departments. **CONSIDERING SUPERIOR
ADVANTAGES FEES ARE LOW.**

For Catalogue and Other Information, Address:

GEORGE M. WATERS, A. M., M. D., Dean, Medical Department.

L. P. BETHEL, D. D. S., Dean, Dental Department.

GEO. H. MATSON, JR., G. PH., Dean, Pharmacy Department.

OHIO MEDICAL UNIVERSITY,
700-716 North Park Street, COLUMBUS, OHIO.

Starling Medical College,

Corner State and 6th Streets,
COLUMBUS, OHIO. 🐼 🐼

*MEDICAL AND SURGICAL CLINICS AT FOUR
WELL EQUIPPED HOSPITALS. 🐼 🐼 🐼 🐼*

WELL EQUIPPED AND THOROUGH IN EVERY DETAIL.
EXCEPTIONAL CLINICAL ADVANTAGES.

CURTIS C. HOWARD, M. Sc., Registrar. STARLING LOVING, M. D.,
LL. D., Dean.

TRANSCRIBER'S NOTES

1. Silently corrected obvious typographical errors and variations in spelling.
2. Retained archaic, non-standard, and uncertain spellings as printed.

*** END OF THE PROJECT GUTENBERG EBOOK THE OHIO
NATURALIST, VOL. II, NO. 2, DECEMBER, 1901 ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.